TR1 Worked Examples: Degrees, Radians and Standard Position

Recording companion for the TR1 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

How much of a half turn is this angle? A half turn is 180 degrees and also π radians, so 180° = π. To go from degrees to radians, multiply by π/180; from radians to degrees, multiply by 180/π. To draw an angle in standard position, put its vertex at the origin, start on the positive x-axis, and turn counterclockwise for a positive angle.

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Example 1: 150 Degrees

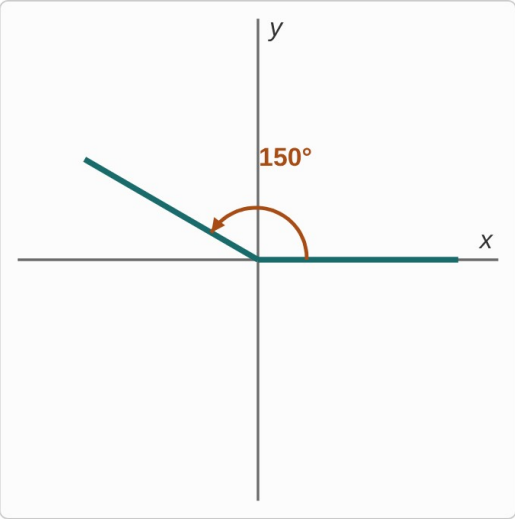


Figure: An angle in standard position: the initial side lies on the positive x-axis, and an arc labeled 150° turns counterclockwise to the terminal side, which points up and to the left, 30° short of the negative x-axis.

$$150°$$

Question

What is this angle in radians?

- **A.** $6\pi/5$
- **B.** $5\pi/6$
- **C.** $5/6$
- **D.** $6/5$

Answer and reasoning

$150° \cdot \pi/180 = 150\pi/180 = 5\pi/6$, dividing top and bottom by 30. A check: 150° is a little less than 180°, and $(5/6)\pi$ is a little less than π. $6\pi/5$ multiplies by $180/150$, upside down, and is more than π; $5/6$ drops the π, which is about 0.83 radians, less than a quarter turn; and $6/5$ makes both slips. ***5π/6*.**

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Example 2: 225 Degrees

$$225°$$

Question

What is this angle in radians?

A. $5\pi/4$
B. $5\pi/8$
C. $4\pi/5$
D. $8\pi/5$

Answer and reasoning

$225 \cdot \pi/180 = 225\pi/180 = 5\pi/4$, dividing by 45. 225° is 45° past 180°, and $5\pi/4$ is $\pi/4$ past π. $5\pi/8$ divides by 360 instead of 180; $4\pi/5$ flips the fraction; and $8\pi/5$ makes both slips. ***5π/4*.**

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Example 3: 270 Degrees

$$270°$$

Question

What is this angle in radians?

A. $2\pi/3$
B. $3\pi/4$
C. $4\pi/3$
D. $3\pi/2$

Answer and reasoning

$270 \cdot \pi/180 = 3\pi/2$, dividing by 90. It is a landmark: three quarter turns, pointing straight down. $2\pi/3$ flips the fraction; $3\pi/4$ divides by 360 instead of 180; and $4\pi/3$ makes both slips. ***3π/2*.**

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Example 4: 330 Degrees

$$330°$$

Question

What is this angle in radians?

A. $11\pi/12$
B. $11\pi/6$
C. $6\pi/11$
D. $12\pi/11$

Answer and reasoning

$330 \cdot \pi/180 = 330\pi/180 = 11\pi/6$, dividing by 30. 330° is 30° short of a full turn, and $11\pi/6$ is $\pi/6$ short of 2π. $11\pi/12$ divides by 360 instead of 180; $6\pi/11$ flips the fraction; and $12\pi/11$ makes both slips. ***11π/6*.**

Example 5: Two Thirds of Pi

$$2\pi/3$$

Question

What is this angle in degrees?

A. 240°
B. 270°
C. 120°
D. about 0.04°

Answer and reasoning

$2\pi/3 \cdot 180/\pi = 360/3 = 120°$. Two thirds of a half turn is 120°. 240° uses $\pi = 360°$, a full turn instead of a half; 270° flips the fraction to $3/2 \cdot 180$; and about 0.04° multiplies by $\pi/180$ instead of $180/\pi$, which gives $(2\pi^2)/540$. **120°.**

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Example 6: Seven Fourths of Pi

$$7\pi/4$$

Question

What is this angle in degrees?

A. 315°
B. 630°
C. about 0.1°
D. about 102.9°

Answer and reasoning

$7\pi/4 \cdot 180/\pi = 7 \cdot 45 = 315°$. Each $\pi/4$ is 45°, and there are 7 of them. 630° uses $\pi = 360°$; about 0.1° multiplies by $\pi/180$, giving $(7\pi^2)/720$; and about 102.9° flips the fraction to $4/7 \cdot 180$. **315°.**

Example 7: Pi

$$\pi$$

Question

CheckIt writes this angle as π. What is it in degrees?

- **A.** 360°
- **B.** 180°
- **C.** about 0.05°
- **D.** about 3.14°

Answer and reasoning

π radians is a half turn: $\pi \cdot 180/\pi = 180°$. 360° is a full turn, 2π; about 0.05° multiplies by $\pi/180$, giving $(\pi^2)/180$; and about 3.14° writes the number π with a degree sign, but π radians is not π degrees. **180°.**

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Example 8: Seven Sixths of Pi

$$7\pi/6$$

Question

What is this angle in degrees?

A. 105°
B. about 0.06°
C. 210°
D. about 154.3°

Answer and reasoning

$7\pi/6 \cdot 180/\pi = 7 \cdot 30 = 210°$. Each $\pi/6$ is 30°. 105° uses $\pi = 90°$, a quarter turn instead of a half; about 0.06° multiplies by $\pi/180$, giving $(7\pi^2)/1080$; and about 154.3° flips the fraction to $6/7 \cdot 180$. **210°.**

Example 9: Read the Angle

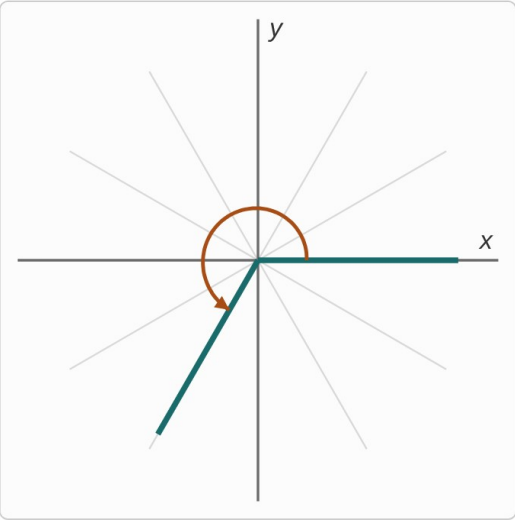


Figure: An angle in standard position, with faint guide lines every 30°. The arc turns counterclockwise from the positive x-axis, past the positive y-axis and the negative x-axis, and stops on the second guide line after the negative x-axis, below and to the left of the origin.

Question

The guide lines are 30° apart. What is the angle, in radians?

- **A.** $2\pi/3$
- **B.** $5\pi/3$
- **C.** $4\pi/3$
- **D.** $7\pi/6$

Answer and reasoning

The arc passes the negative x-axis (180°) and goes 2 guide lines farther, $180°+60°=240°$. In radians, $240\cdot\pi/180=4\pi/3$. $2\pi/3$ is 120°, the mirror image across the x-axis, above it instead of below; $5\pi/3$ is 300°, the mirror image across the y-axis; and $7\pi/6$ is 210°, one guide line short. ***4π/3*.**

Example 10: Which Quadrant

$$5\pi/4$$

Question

In which quadrant does the terminal side of this angle lie?

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Answer and reasoning

Compare with the landmarks: $\pi=4\pi/4$ and $3\pi/2=6\pi/4$. So $5\pi/4$ is between a half turn and three quarter turns, in Quadrant III, 45° past the negative x-axis. Quadrant I would need less than $\pi/2$; Quadrant II, between $\pi/2$ and π; and Quadrant IV, between $3\pi/2$ and 2π. **Quadrant III.**

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Example 11: On an Axis

$$270°$$

Question

Where is the terminal side of this angle in standard position?

A. Along the positive y-axis
B. Along the negative y-axis
C. Along the negative x-axis
D. Inside the third quadrant

Answer and reasoning

Counterclockwise from the positive x-axis: 90° points up, 180° points left, and 270° points down. So the terminal side lies along the negative y-axis. An angle on an axis is in no quadrant. The positive y-axis is 90°; the negative x-axis is 180°; and the third quadrant is strictly between 180° and 270°. **Along the negative y-axis.**

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Example 12: Describe the Turn

$$5\pi/3$$

Question

Which describes how to draw this angle in standard position?

A. Turn clockwise 300° from the positive x-axis
B. Turn counterclockwise 5/3 of a full turn
C. Turn counterclockwise 300° from the positive x-axis
D. Turn counterclockwise 300° from the positive y-axis

Answer and reasoning

$5\pi/3 \cdot 180/\pi = 300°$. A positive angle in standard position starts on the positive x-axis and turns counterclockwise, so turn 300° counterclockwise, ending in Quadrant IV. Turning clockwise 300° would be the angle $-5\pi/3$; $5\pi/3$ is $5/6$ of a full turn, 2π, not $5/3$ of one; and standard position always starts on the positive x-axis, not the y-axis. **Turn counterclockwise 300° from the positive x-axis.**

Example 13: Slip 1

$$240°\cdot 180/\pi = 43200/\pi$$

Question

A student converted 240° to radians. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** Use π/180: 240π/180 = 4π/3.
- **C.** Divide by 360 instead, which gives 240/360 = 2/3.
- **D.** Use π = 3.14 first: 240 × 3.14 = 753.6.

Answer and reasoning

To go to radians, the degrees must cancel, so multiply by $\pi/180$: $240\pi/180 = 4\pi/3$. The student's $43200/\pi \approx 13{,}751$ radians would be over 2,000 full turns. Dividing by 360 gives $2/3$, the fraction of a full turn, not the angle in radians; and using $\pi = 3.14$ first, 240×3.14, is not the conversion at all. ***4π/3*.**

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Example 14: Slip 2

$$5\pi/6 \cdot 180/\pi = 900/6 = 150°$$

Question

A student converted 5π/6 to degrees. What is the mistake, if any?

A. There is no mistake.
B. Multiply by π/180 instead, to get 5π²/1080.
C. Degrees need π too: the answer is 150π°.
D. 5π/6 is in Quadrant III, so it is 210°.

Answer and reasoning

Multiplying by $180/\pi$ cancels the π: $(5 \cdot 180)/6 = 150°$. The student is right. Multiplying by $\pi/180$ goes the wrong way and leaves π^2; the π cancels, so degrees have none; and $5\pi/6$ is less than π, so it is in Quadrant II, not III. **There is no mistake.**

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Example 15: Slip 3

$$315° \cdot \pi/360 = 7\pi/8$$

Question

A student converted 315° to radians. What is the mistake, if any?

A. There is no mistake.
B. The π should not be there: the answer is 7/8.
C. Use 180, not 360: 315π/180 = 7π/4.
D. Flip the fraction: the answer is 8π/7.

Answer and reasoning

π radians is 180°, not 360°, so the factor is $\pi/180$: $315\pi/180 = 7\pi/4$. A check: 315° is almost a full turn, and $7\pi/4$ is almost 2π, but $7\pi/8$ is less than a half turn. Radians for these angles keep the π; and the student's fraction is not upside down, it uses the wrong half-turn value. ***7π/4*.**

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Example 16: Slip 4

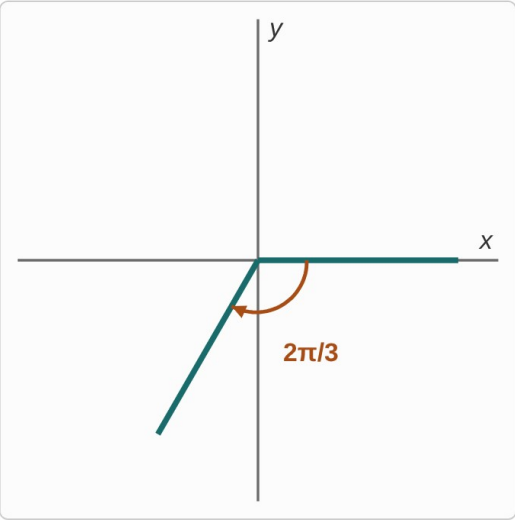


Figure: A student's drawing of an angle labeled 2π/3. The arc starts on the positive x-axis and turns clockwise, downward, ending with the terminal side below and to the left of the origin.

$$2\pi/3$$

Question

A student drew this angle in standard position. What is the mistake, if any?

A. There is no mistake.
B. The angle should start from the positive y-axis instead.
C. 2π/3 is more than half a turn, so it ends below the x-axis.
D. It should turn counterclockwise, into Quadrant II.

Answer and reasoning

$2\pi/3 = 120°$, a positive angle, so it turns counterclockwise from the positive x-axis and ends in Quadrant II, above the axis. The student turned 120° clockwise, which is the angle $-2\pi/3$. Standard position starts on the positive x-axis; and $2\pi/3$ is less than π, so it is less than half a turn. **Turn counterclockwise, into Quadrant II.**

Example 17: Slip 5

$$3\pi/4=(3(3.14))/4\approx 2.36°$$

Question

A student converted 3π/4 to degrees. What is the mistake, if any?

A. There is no mistake.
B. That is radians; times 180/π, it is 135°.
C. Use π = 3 instead, so the answer is 2.25°.
D. Multiply by π/180, so the answer is about 0.04°.

Answer and reasoning

About 2.36 is $3\pi/4$ as a number of radians, not degrees. To get degrees, multiply by $180/\pi$: $3\pi/4 \cdot 180/\pi = 3 \cdot 45 = 135°$. A different value for π still leaves the answer in radians; and multiplying by $\pi/180$ goes the wrong way. **135°.**

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